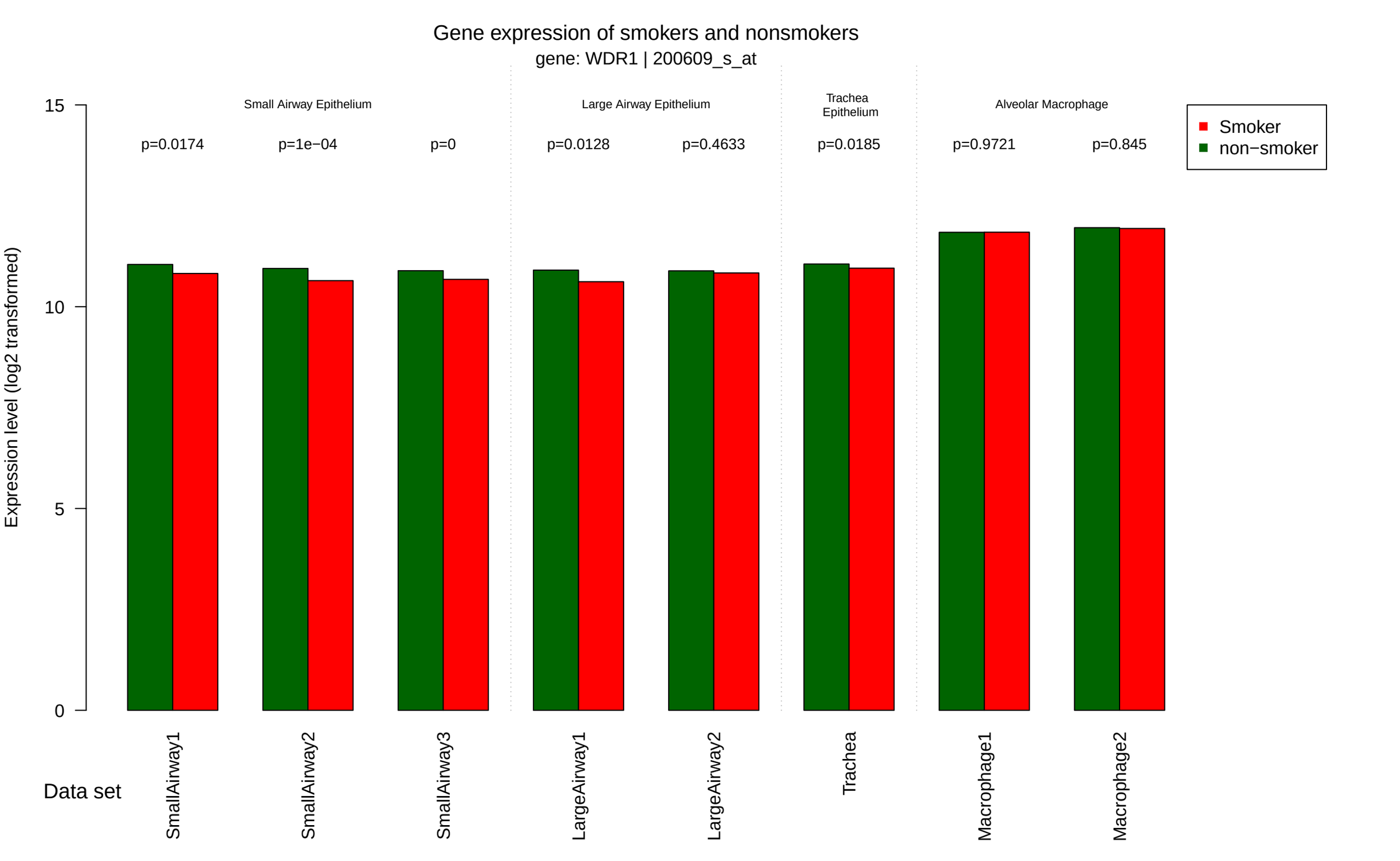
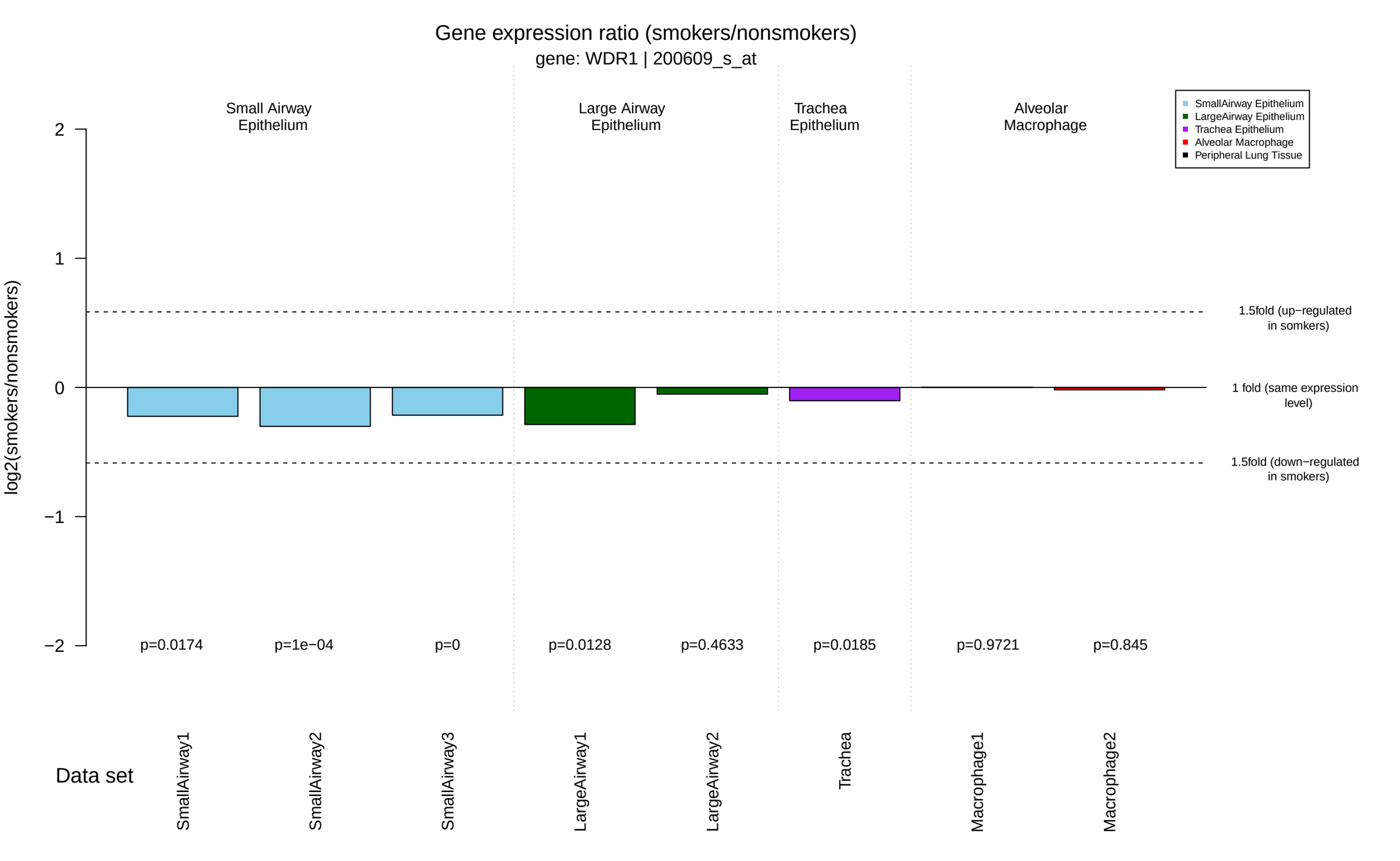
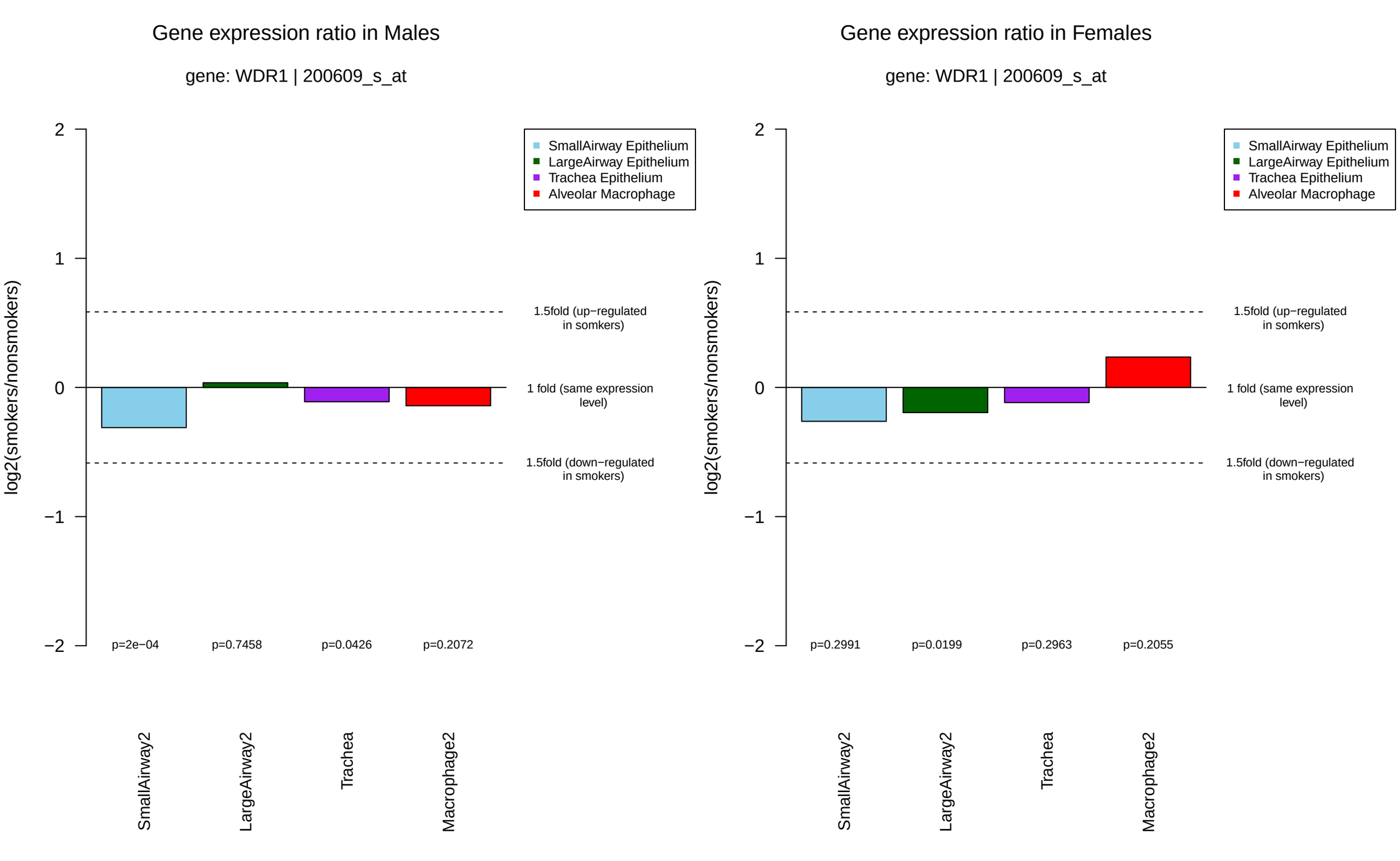
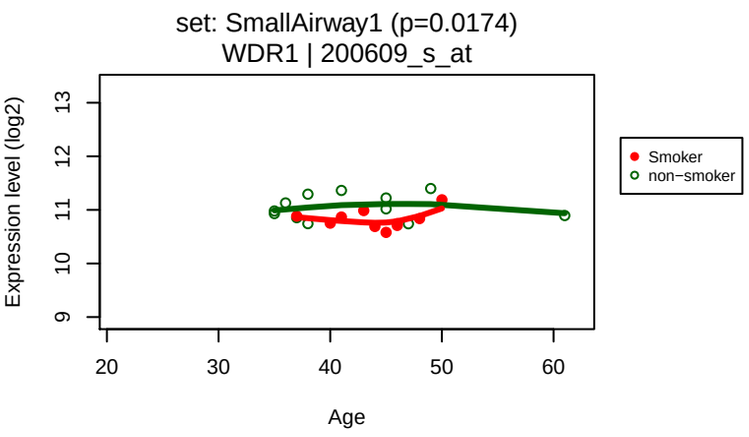
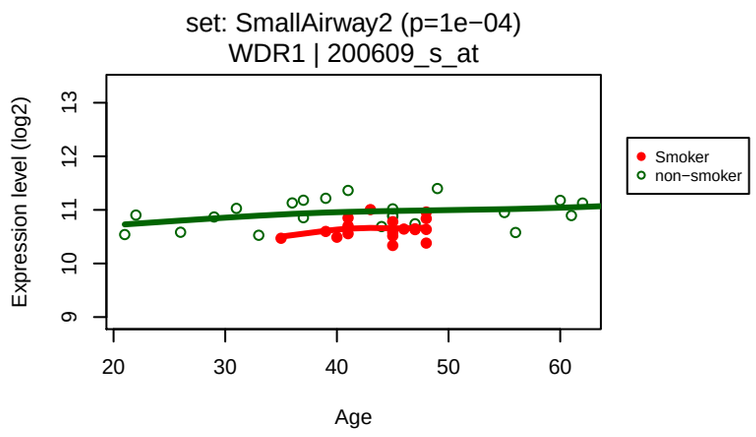
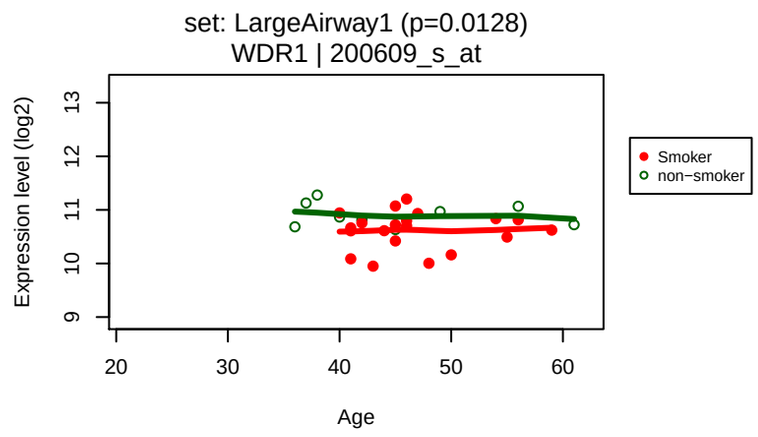

Gene expression of smokers and nonsmokers
gene: WDR1 | 200609_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0174
p=1e−04
p=0
p=0.0128
p=0.4633
p=0.0185
p=0.9721
p=0.845
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

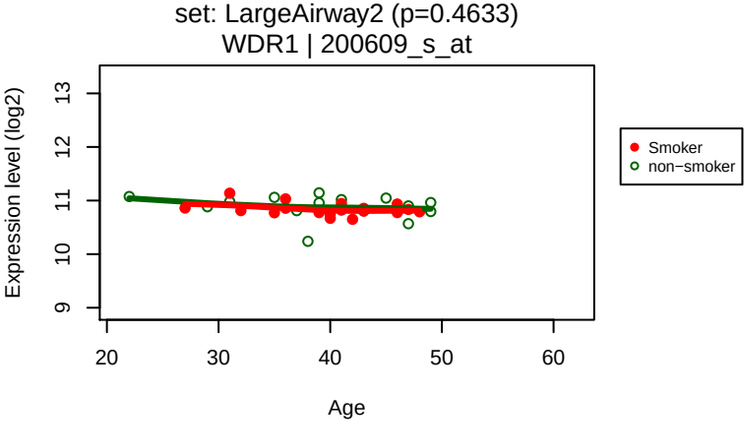

Gene expression ratio (smokers/nonsmokers)
gene: WDR1 | 200609_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0174
p=1e−04
p=0
p=0.0128
p=0.4633
p=0.0185
p=0.9721
p=0.845
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

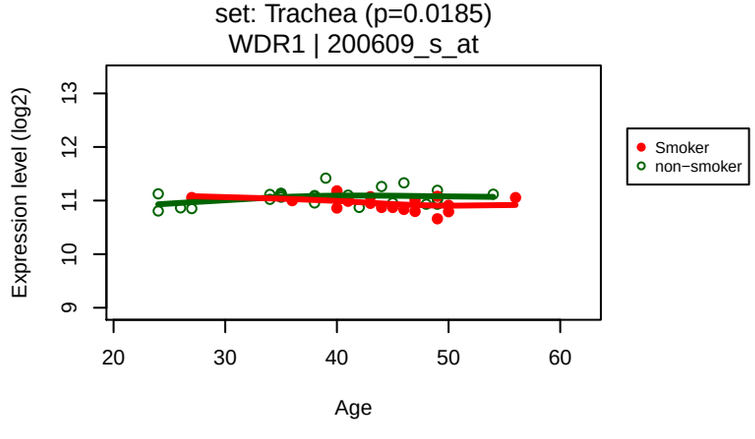

Gene expression ratio in Males
gene: WDR1 | 200609_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=2e−04
p=0.7458
p=0.0426
p=0.2072
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WDR1 | 200609_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2991
p=0.0199
p=0.2963
p=0.2055
SmallAirway2
LargeAirway2
Trachea
Macrophage2

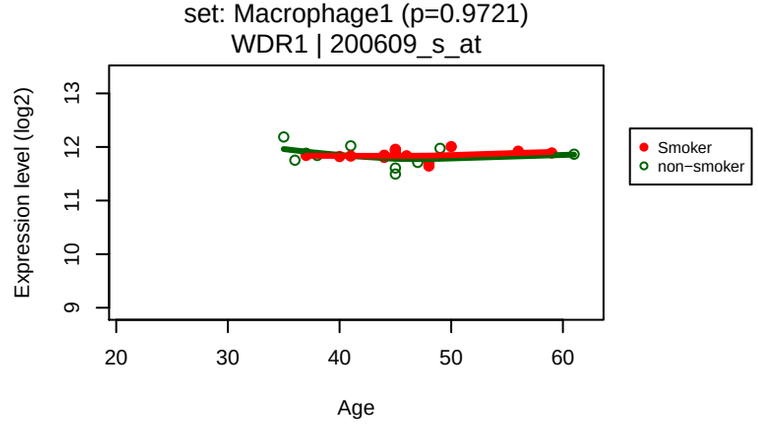

set: SmallAirway1 (p=0.0174)
WDR1 | 200609_s_at
Expression level (log2)
Age
Smoker
non−smoker

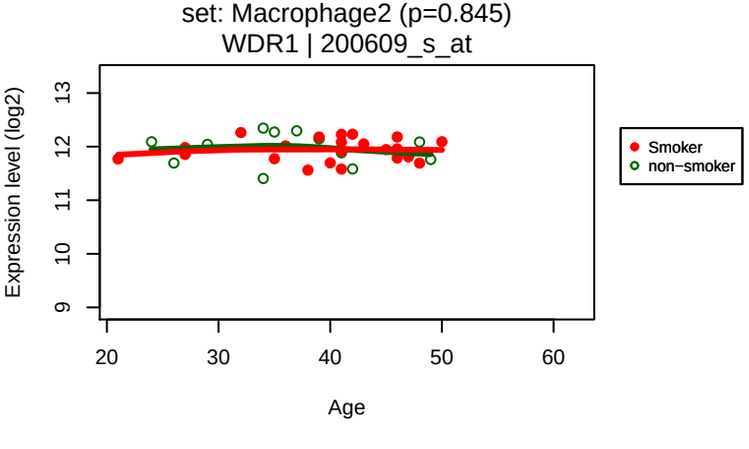

set: SmallAirway2 (p=1e−04)
WDR1 | 200609_s_at
Expression level (log2)
Age
Smoker
non−smoker

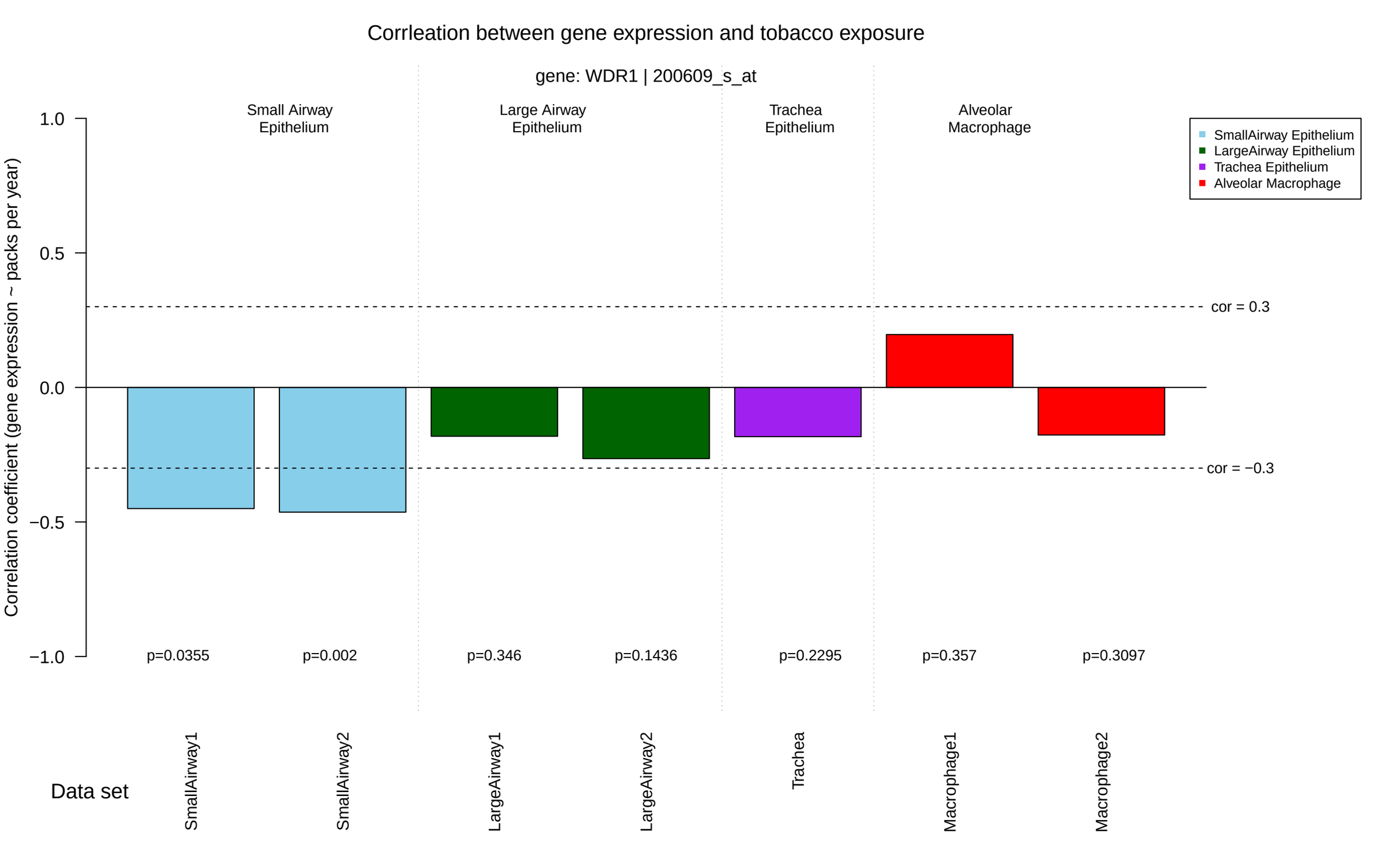

set: LargeAirway1 (p=0.0128)
WDR1 | 200609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4633)
WDR1 | 200609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0185)
WDR1 | 200609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9721)
WDR1 | 200609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.845)
WDR1 | 200609_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WDR1 | 200609_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0355
p=0.002
p=0.346
p=0.1436
p=0.2295
p=0.357
p=0.3097
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2